ADC Section 3

2025年11月11日 星期二　09:35

ADC S3.

7 November, 2025

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Reference, ADC section 3

Aim: Test for 1st and 2nd order filters.

Procedure:

- Before the lab.

I am allocated to build a: high-pass filter with 7300Hz corner frequency with $\zeta = 1$.

For high-pass: $H = \frac{Y}{X} = \frac{LC(jw)^2}{Rjwc + 1 + LC(jw)^2}$

$\therefore w_c = \sqrt{\frac{c}{a}} = \sqrt{\frac{1}{LC}} = 2\pi f$

$\therefore \sqrt{\frac{1}{LC}} = 2\pi \times 7300$ ①.

$\zeta = \frac{b}{2aw_c} = \frac{RC}{2LC\sqrt{\frac{1}{LC}}} = \frac{1}{2}\frac{R}{L}\cdot\sqrt{LC} = \frac{1}{2}R\sqrt{\frac{C}{L}}$

$\therefore \frac{1}{2}R\sqrt{\frac{C}{L}} = 1$ ②.

$\therefore$ I choose 1 mH inductor.

$\sqrt{\frac{1}{1\times10^{-3}\times C}} = 2\pi \times 7300$

$C = 4.753\times10^{-7}$ F ≈ 470 nF.

$\frac{1}{2}R\sqrt{\frac{4.753\times10^{-7}}{1\times10^{-3}}} = 1$

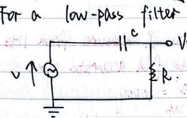
$\therefore R = 91.74\Omega \approx 91\Omega$

$\therefore$ $L = 1$mH
$C = 470$nF
$R = 91\Omega$

- First-order filter.

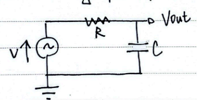
For each group of high-pass and low-pass filter, we measure impedance of capacitor across different frequency, to find out overall trend of $|Z|$ across frequency, and find the corner frequency.

For a low-pass filter.



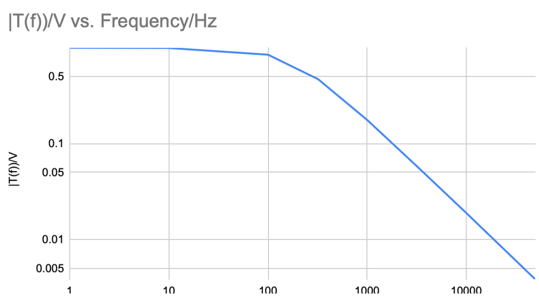
Resistance/Ohm	Capacitance/uF	Lowpass Frequency/Hz	DeltaT/s	Vout/V	Vin/V	Arg(T(f))	|T(f))/V
1000	1	1	0.00242	1.72	1.73	0.015205308	0.994219653
1000	1	10	0.00104	1.79	1.81	0.065345127	0.988950276
1000	1	100	0.00083664	1.45	1.72	0.525676416	0.843023256
1000	1	320	0.00083232	0.77718	1.66	1.673478654	0.468180723
1000	1	1000	0.0002172	0.29232	1.66	1.364707849	0.176096386
1000	1	3200	0.0000739	0.09492	1.65	1.485847661	0.057527273
1000	1	10000	0.00002436	0.03099	1.64	1.530583941	0.018896341
1000	1	50000	0.00000492	0.00573	1.5	1.545663586	0.00382

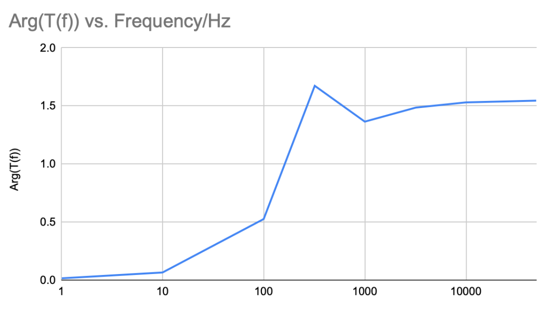
For a high-pass filter.

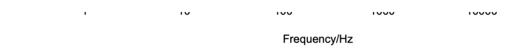


Resistance/Ohm	Capacitance/uF	Highpass Frequency/Hz	DeltaT/s	Vout/V	Vin/V	Arg(T(f))	|T(f))/V
1000	1	1	0.22772	0.01133	1.73	1.430806958	0.006549133
1000	1	10	0.02333	0.11102	1.81	1.465867132	0.061337017
1000	1	100	0.00165	0.88272	1.71	1.036725576	0.516210526
1000	1	320	0.000244	1.46	1.66	0.490591109	0.879518072
1000	1	1000	0.0000284	1.62	1.64	0.178442463	0.987804878
1000	1	3200	0.0000028	1.64	1.65	0.05629734	0.993939394
1000	1	10000	0.00000038	1.65	1.66	0.023876104	0.993975904
1000	1	50000	0.00000001	1.51	1.51	0.003141593	1

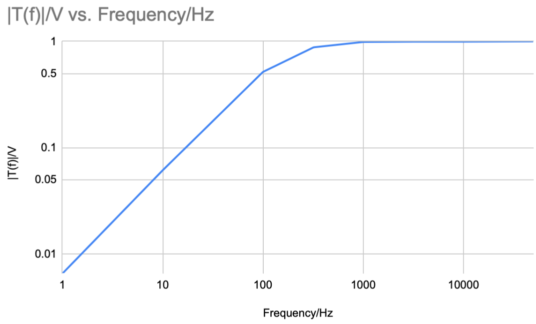
For low-pass:

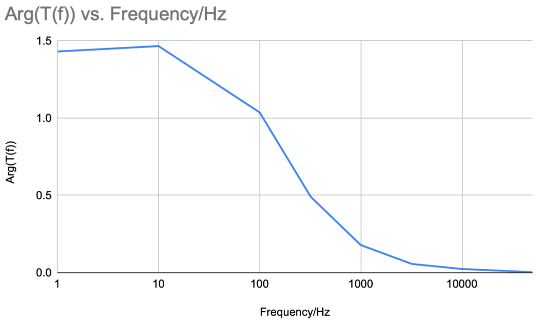






For high-pass:





They work well and allign with the low-pass and high-pass filters' functions.

- Resonant networks.

Using same method of CH1, CH2, to measure impedances of series and parallel LC network, and find resonant frequency.

Note: At first, I didn't see that we have to predict Wo. which is $WL - \frac{1}{WC} = 0$

$W_0 = \sqrt{\frac{1}{LC}}$, so I measure from 1Hz to 10kHz, which makes it impossible to find accurate Wo.

then I do $W_0 = \sqrt{\frac{1}{100\times10^{-9}\times100\times10^{-3}}} = 10000$

$\therefore f = \frac{W_0}{2\pi} = 1591.5$ Hz,

so measure from 1000 ~ 2000 Hz.

Series network:

resonance 1				
Resistance/Ohm	Vz/V	VR/V	impedence/(	Frequency/Hz
1000	1.72	1.11	1549.5495	1000
1000	1.71	1.27	1346.4567	1200
1000	1.71	1.36	1257.3529	1400
1000	1.7	1.38	1231.8841	1500
1000	1.66	1.35	1229.6296	1600
1000	1.66	1.34	1238.806	1620
1000	1.66	1.35	1229.6296	1640
1000	1.66	1.35	1229.6296	1680
1000	1.66	1.34	1238.806	1700
1000	1.71	1.37	1248.1752	1800
1000	1.72	1.32	1303.0303	2000
1000	1.72	1.25	1376	2200

Parallel network:

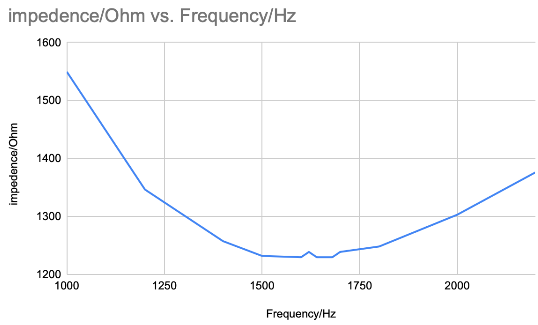
Resonance 2				
Resistance/Ohm	Vz/V	VR/V	impedence/Ohm	Frequency/Hz
5000	1.72	1.54	5584.415584	1000
5000	1.72	1.42	6056.338028	1200
5000	1.72	1.19	7226.890756	1400
5000	1.72	1.02	8431.372549	1500
5000	1.72	0.90228	9531.409319	1600
5000	1.72	0.89855	9570.97546	1620
5000	1.72	0.90422	9510.959722	1640
5000	1.72	0.93232	9224.300669	1680
5000	1.72	1.12	7678.571429	1800
5000	1.72	1.4	6142.857143	2000
5000	1.72	1.54	5584.415584	2200

Wrong version:

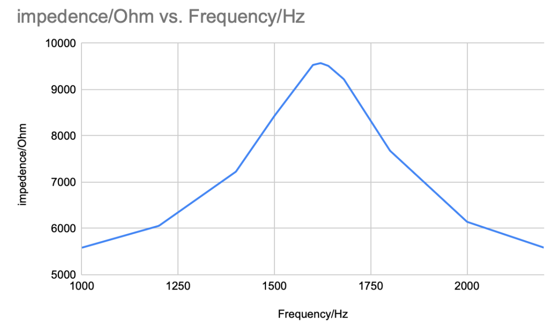
Wrong version				
resonance 1				
Resistance/Ohm	Vz/V	VR/V	impedence/Ohm	Frequency/Hz
1000	1.39	0.98022	1418.049009	1
10000	1.8	0.11265	159786.9507	10
10000	1.74	0.34354	50649.12383	32
10000	1.46	0.89991	16223.84461	100
10000	0.18376	1.67	1100.359281	1000
10000	0.88317	1.45	6090.827586	10000
10000	1.51	0.68687	21983.7815	32000
10000	1.53	0.37534	40763.04151	50000

resonance 2				
Resistance/Ohm	Vz/V	VR/V	impedence/Ohm	Frequency/Hz
1000	0.29404	1.37	214.6277372	1
1000	0.30505	1.42	214.8239437	3.2
1000	0.30794	1.43	215.3426573	10
1000	0.30411	1.41	215.6808511	32
1000	0.30854	1.37	225.2116788	100
1000	0.40783	1.33	306.6390977	320
1000	0.99412	0.94008	1057.484469	1000
1000	0.29095	1.63	178.4969325	10000

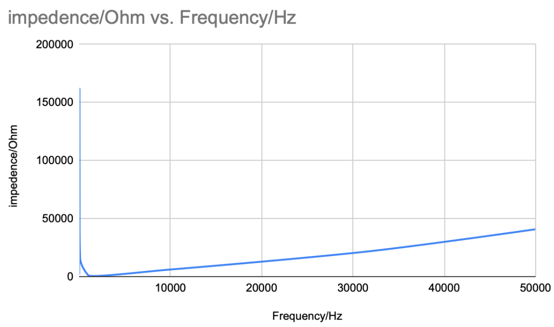
Series network: Can observe that the lowest impedance, which is at the resonant frequency is at near 1600Hz. The graph alligns with the data.

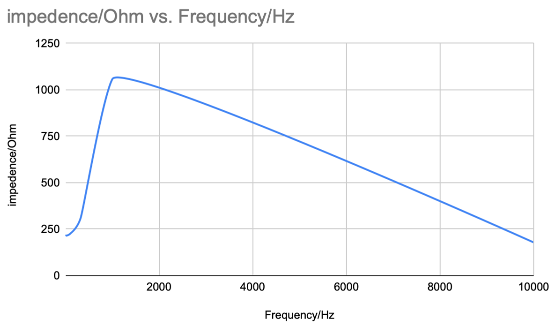


Parallel network: Can observe that the highest impedance, which is at the resonant frequency is at near 1600Hz. The graph alligns with the data.



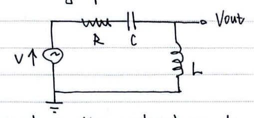
Wrong version:



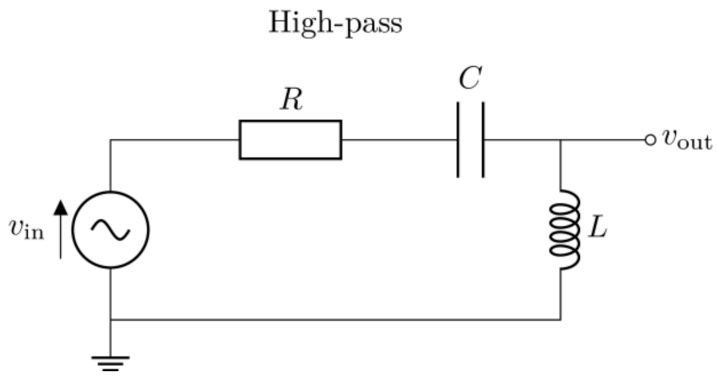


X and Y axis is too large so cannot read numbers accurately.

· Challenge: test my second-order filter.
I have a high-pass.

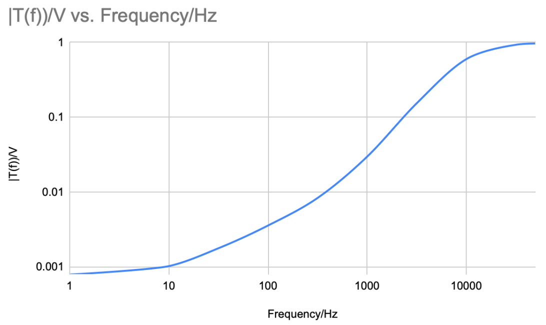


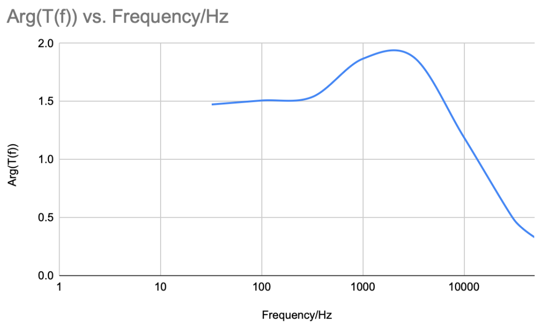
It's trend is like predicted in the LTSpice, and has the function of high-pass filter.
Small discrepancies may because of:
· Inideal behavior of components and uncertainty.
· Parasitic resistance.



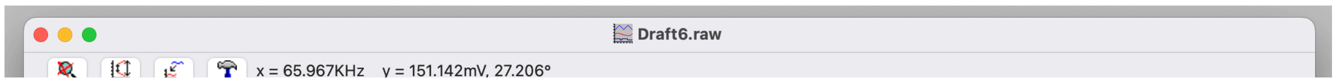
Second-order highpass

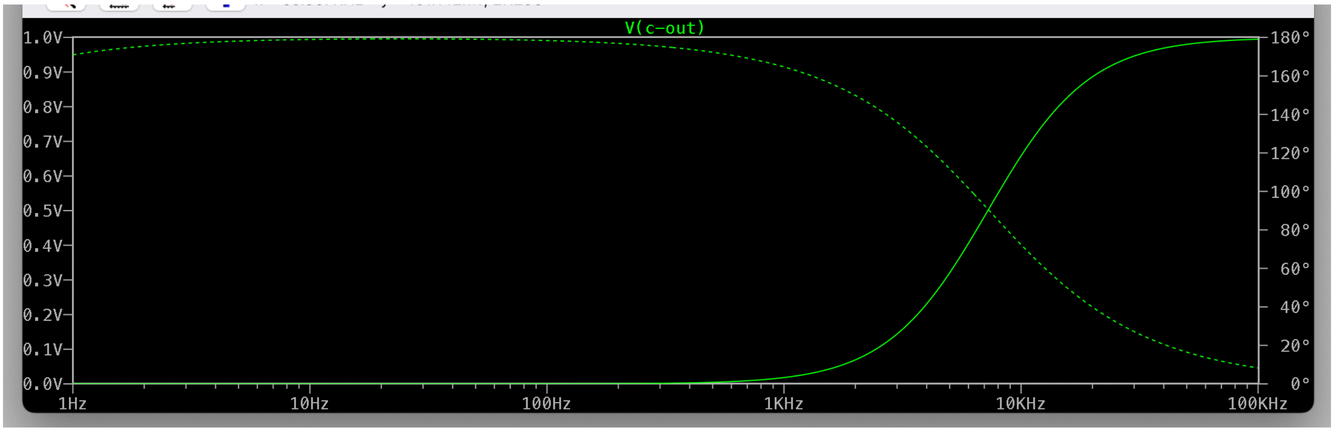
Frequency/Hz	DeltaT/s	Vout/V	Vin/V	Arg(T(f))	|T(f))/V
1	Too small	0.00137	1.73	/	0.000791908
10	Too small	0.00186	1.81	/	0.001027624
32	0.00733	0.00319	1.78	1.473783946	0.001792135
100	0.0024	0.00623	1.74	1.507964474	0.00358046
320	0.000766	0.01455	1.72	1.540134382	0.008459302
1000	0.0002972	0.0492	1.65	1.867362673	0.029818182
3200	0.0000935	0.20957	1.34	1.879929044	0.156395522
10000	0.00001896	0.69869	1.17	1.191291934	0.59717094
32000	0.00000234	1.36	1.46	0.470484916	0.931506849
50000	0.00000105	1.44	1.49	0.329867229	0.966442953

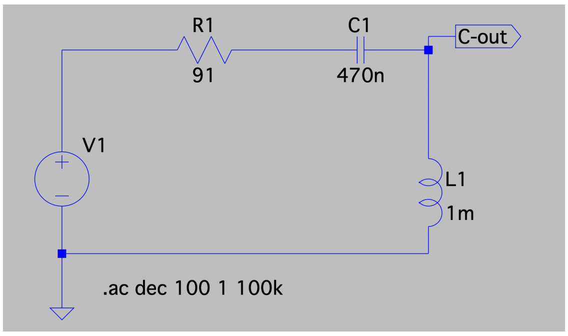




Simulation on Ltspice:




V(c-out)
1.0V
0.9V
0.8V
0.7V
0.6V
0.5V
0.4V
0.3V
0.2V
0.1V
0.0V
180°
160°
140°
120°
100°
80°
60°
40°
20°
0°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz


R1
91
C1
470n
C-out
V1
L1
1m
.ac dec 100 1 100k